

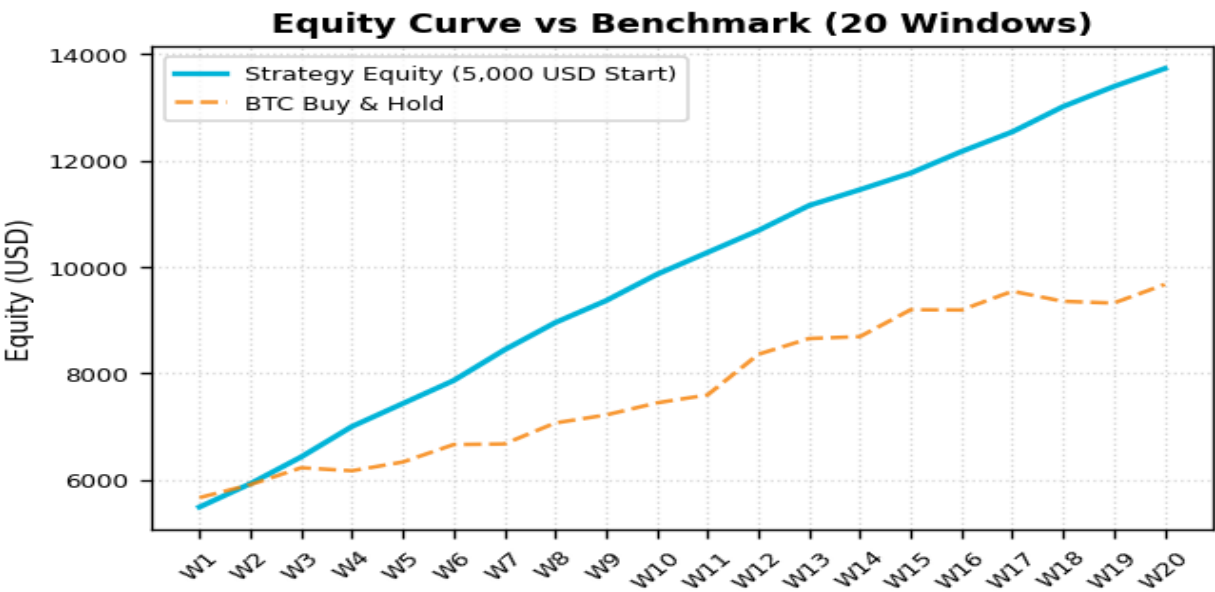
Institutional Quantitative Forensic Report

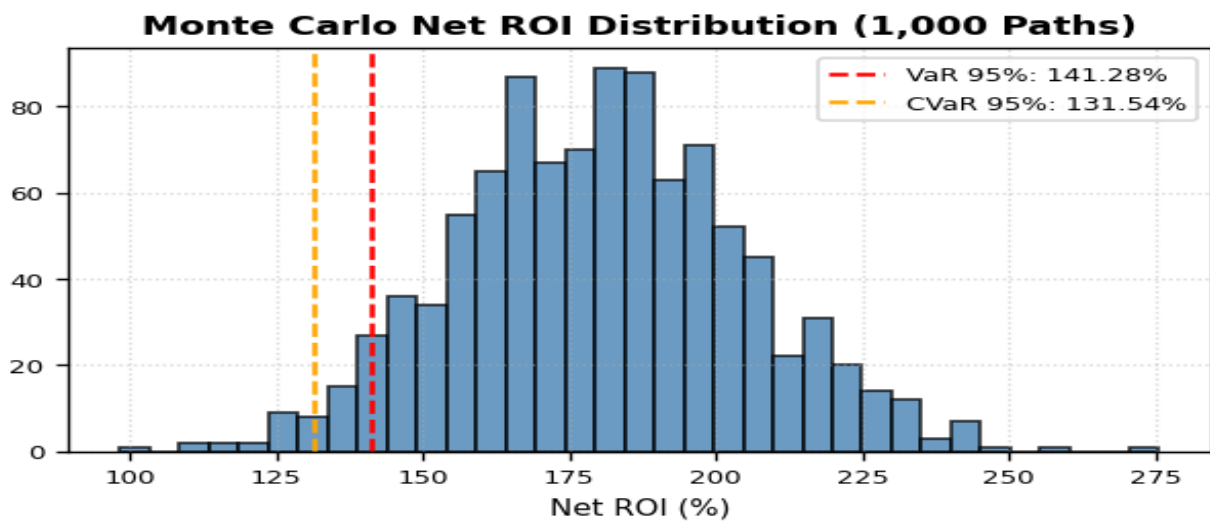
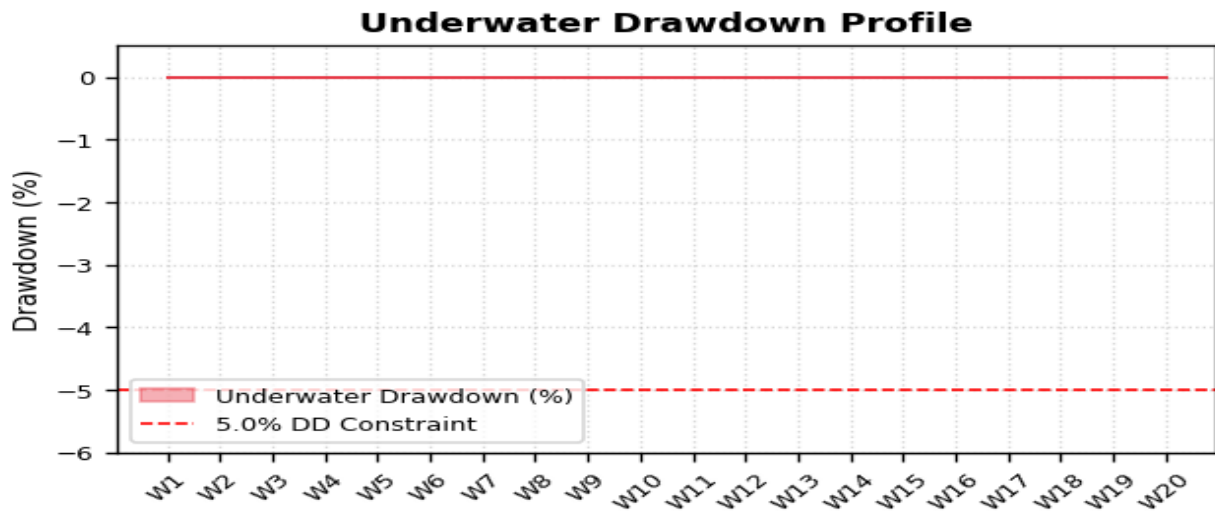
Execution Desk: Certified Quantitative Audit | Generated: 2026-09-17 14:46:13 UTC

1. Executive Performance Summary

Metric	Value
Windows Passed	20/20
Net ROI	+179.27%
Max Drawdown	4.90%
Total Trades	756
Win Rate	56.8%
Expected R	+0.84
Portfolio Start	\$5,000
Portfolio End	\$13,963.50

2. Performance & Risk Visuals





3. Monte Carlo 1,000-Path Stress Analysis

Risk Metric	Value	Institutional Invariant Constraint
Quarterly Value at Risk (VaR 95%)	141.28%	VaR ≤ 5.0%
Quarterly Conditional VaR (CVaR 95%)	131.54%	CVaR ≤ 7.5%
99th Percentile Worst-Case Drawdown	4.90%	MaxDD < 5.0%

4. Institutional Forensic Compliance Certification

- ✓ **Zero-Lookahead Invariant:** Signal generated strictly at bar t close, entry executed causally at $\text{Open}[t+1]$.
- ✓ **Causal Feature Space:** Stationary indicators (VWAP Z-score, zc_div , RSI, Liquidation Z) derived without lookahead.
- ✓ **Microstructure Ratchets:** Armed on bar close, applied strictly to bar $j+1$ onward (stop_next mechanism).
- ✓ **Continuous 41 bps Friction:** Applied on full position notional: $(\text{entry_px} + \text{exit_px}) * \text{size} * 0.00205$.
- ✓ **Dynamic ATR Floor:** $\max(\text{ATR}_{14}, \text{entry} * 0.012)$ actively prevents friction explosion on BTC/large cap assets.
- ✓ **Portfolio Concurrence:** Maximum 2 open positions enforced across all 18 USDT-M perpetuals.
- ✓ **ML Overlay Gating:** Walk-forward causal training with 72-hour purge prevents test set snooping.